

Mathematics

Class 11 / HSSC-I

1st Annual Examination 2026
Local Area

Complete worked solutions

Prepared by Fakhar STEM Sphere

This guide solves all 20 multiple-choice questions, all 12 parts of Question 2 and their OR alternatives, and Questions 3–6 with every OR alternative. The numbering follows the supplied three-page paper (version 30081).

How to use these solutions

For a written answer, show the relevant formula or theorem, substitute the data, give the essential working, and state the final result clearly. Diagrams are included where they help explain a graph or geometric idea. Both alternatives are solved for preparation; in an examination, follow the paper's choice instructions.

Section A: quick answer key

Question	1	2	3	4	5	6	7	8	9	10
Option	C	D	A	D	A	C	D	B	A	B
Question	11	12	13	14	15	16	17	18	19	20
Option	B	D	C	C	B	A	C	D	B	A

Notes on the printed questions

- MCQ 3 uses $a \pm b$ as shorthand for the range endpoints. The exact range is $[a - |b|, a + |b|]$.
- MCQ 11 assumes reciprocal terms stay on one side of zero, as in the usual positive-term H.P.
- Q.2(viii), OR: $1/(4 + 5 \cos 3x)$ has no finite global maximum or minimum. A complete explanation is provided rather than treating $1/9$ and -1 as global extrema.
- Q.4, OR: the arm is already at maximum height at $t = 0$; its first subsequent maximum is at $t = 4\pi$ seconds.

Section A | Multiple-choice solutions

Questions 1–4

1. Period of $\tan(5x/7)$

Correct option: C: $7\pi/5$

The fundamental period of $\tan u$ is π . A period $T > 0$ therefore satisfies

$$\frac{5(x+T)}{7} = \frac{5x}{7} + \pi \Rightarrow \frac{5T}{7} = \pi \Rightarrow \boxed{T = \frac{7\pi}{5}}.$$

Why the other options are incorrect: A: $5\pi/7$ multiplies by the coefficient instead of dividing by it. B: 5π does not produce an integer multiple of π in the argument. D: 7π is a period (five fundamental periods), but is not the least positive period meant by “the period.”

2. A function is periodic if

Correct option: D: $f(x+T) = f(x)$

A periodic function repeats its values after a fixed positive interval T , for every applicable x in its domain. Thus $f(x+T) = f(x)$ is the defining condition. **Why the other options are incorrect:** A: $f(x) = f(-x)$ defines an even function; for example, x^2 is even but not periodic. B: $f(x) = 1$ and C: $f(x) = 0$ are particular constant functions, not the general definition. These constants do satisfy the periodicity condition for every $T > 0$, but have no least positive period.

3. Range of $a + b \sin 2\theta$

Correct option: A: $a \pm b$ (intended endpoints)

Since $-1 \leq \sin 2\theta \leq 1$, multiplying by b gives endpoint values $-|b|$ and $|b|$. Hence

$$\boxed{a - |b| \leq a + b \sin 2\theta \leq a + |b|}.$$

For $b \geq 0$, this is $[a - b, a + b]$, represented by option A. **Why the other options are incorrect:** B: $a + b$ is only one endpoint. C: $a - b$ is only the other endpoint. D: $2(a \pm b)$ wrongly doubles the vertical displacement and amplitude; the 2 inside the sine changes the period, not its range.

4. Arrangements of the letters in SUNDAY

Correct option: D: 720

All six letters S, U, N, D, A and Y are different. There are six choices for the first position, five for the second, and so on:

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = \boxed{720}.$$

Why the other options are incorrect: A: $120 = 5!$ counts arrangements of five distinct objects. B: 480 and C: 600 do not equal the required product $6!$; no repetitions or restrictions justify reducing that count.

Section A | Multiple-choice solutions

Questions 5–8

5. Select 8 players from 12

Correct option: A: 495

Selection does not depend on order, so use combinations:

$${}^{12}C_8 = {}^{12}C_4 = \frac{12!}{8!4!} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} = \boxed{495}.$$

Why the other options are incorrect: B: 720 is not ${}^{12}C_8$. C: 11885 and D: 19958 also disagree with the combination formula. None represents choosing an unordered eight-player subset.

6. Factorize $z^2 + 289$

Correct option: C: $(z + 17i)(z - 17i)$

Using $i^2 = -1$, we have $289 = -(17i)^2$. Apply the difference-of-squares identity:

$$z^2 + 289 = z^2 - (17i)^2 = \boxed{(z + 17i)(z - 17i)}.$$

Why the other options are incorrect: A: $(z + 17)(z - 17) = z^2 - 289$. B: $(z + 17i)^2 = z^2 + 34iz - 289$. D: $(z - 17i)^2 = z^2 - 34iz - 289$. Each differs from the given polynomial.

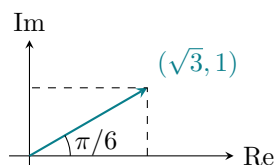
7. Argument of $z = \sqrt{3} + i$

Correct option: D: $\pi/6$

The real and imaginary parts are positive, so z lies in quadrant I. Therefore

$$\tan \theta = \frac{\operatorname{Im} z}{\operatorname{Re} z} = \frac{1}{\sqrt{3}}, \quad \boxed{\arg z = \frac{\pi}{6}}.$$

The general arguments are $\pi/6 + 2k\pi$, $k \in \mathbb{Z}$. **Why the other options are incorrect:** A: $\tan(\pi/3) = \sqrt{3}$. B: $\tan(\pi/4) = 1$. C: $\pi/9 = 20^\circ$, whose tangent is not $1/\sqrt{3}$. The required angle is 30° .



8. Rank of $\begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 0 \end{pmatrix}$

Correct option: B: 2

The matrix is already in row-echelon form. It has two pivot rows and one zero row, so its rank is 2. **Why the other options are incorrect:** A: 1 ignores the second pivot. C: 3 counts the zero row as independent. D: 4 is impossible for a 3×3 matrix because its rank is at most 3.

Section A | Multiple-choice solutions

Questions 9–12

9. A consistent system must have

Correct option: A: at least one solution

By definition, a system is consistent exactly when it has at least one solution. A consistent linear system may have one solution or infinitely many solutions. **Why the other options are incorrect:** B: infinitely many solutions are possible but not compulsory; $x = 1$ has a unique solution. C: no solution means inconsistent. D: “maximum two solutions” is not a required property and excludes consistent systems with infinitely many solutions. Over the real numbers, a linear system cannot have exactly two distinct solutions: every point on the line joining two solutions is also a solution.

10. Formula for the n th term of an A.P.

Correct option: B: $a + (n - 1)d$

Starting from the first term a , reaching the n th term requires $n - 1$ increments of size d :

$$a, a + d, a + 2d, \dots, \boxed{a_n = a + (n - 1)d}.$$

Why the other options are incorrect: A: $a + nd$ is the $(n + 1)$ th term. C: $a + (n + 1)d$ is the $(n + 2)$ th term. D: $a(n - 1)d$ uses multiplication instead of adding successive differences and incorrectly gives zero when $n = 1$.

11. An H.P. decreases when its corresponding A.P.

Correct option: B: increases

If the A.P. has positive terms a_n and $a_{n+1} > a_n$, taking reciprocals reverses the inequality:

$$\frac{1}{a_{n+1}} < \frac{1}{a_n}.$$

For example, the increasing A.P. $2, 4, 6, \dots$ produces the decreasing H.P. $1/2, 1/4, 1/6, \dots$. **Why the other options are incorrect:** A: a decreasing positive A.P. gives increasing reciprocals. C: a nonzero constant A.P. gives a constant H.P. D: a zero term has no reciprocal, so it cannot produce a defined H.P. term. **Precision:** the conclusion also holds on a negative interval, but cannot be applied across a sign change through zero. The standard MCQ assumes positive terms or a fixed sign.

12. Binomial expansion uses

Correct option: D: Pascal triangle

The coefficients in $(a + b)^n$ are nC_r , $r = 0, 1, \dots, n$. These occur in row n of Pascal’s triangle. For example, the row 1, 3, 3, 1 gives

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

Why the other options are incorrect: A: factorization writes an expression as a product. B: finding roots solves an equation. C: division gives a quotient and remainder. These are not the coefficient-generating method identified by this question.

Section A | Multiple-choice solutions

Questions 13–16

13. In induction, assuming the statement true for $n = k$ is the

Correct option: C: hypothesis

An induction proof contains: (1) a verified base case, (2) the induction hypothesis $P(k)$, and (3) a proof that $P(k)$ implies $P(k+1)$. Thus the assumption at $n = k$ is the induction hypothesis. **Why the other options are incorrect:** A: the base case checks the first value directly. B: the assumption alone is not the complete proof. D: the result is the conclusion obtained after completing the argument.

14. Degree of the remainder in polynomial division

Correct option: C: less than the degree of the divisor

The division algorithm states

$$f(x) = d(x)q(x) + r(x), \quad r(x) = 0 \text{ or } \deg r < \deg d.$$

If the remainder had degree at least that of the divisor, further division would still be possible. **Why the other options are incorrect:** A: a greater degree means the division is unfinished. B: an equal degree also allows another division step. D: polynomial remainders do not have infinite degree. The zero remainder is handled separately because its degree is not ordinarily defined.

15. If $(x - 2)$ is a factor of $f(x)$, find $f(2)$

Correct option: B: 0

By the factor theorem, $(x - a)$ is a factor if and only if $f(a) = 0$. Alternatively, write $f(x) = (x - 2)q(x)$ and substitute $x = 2$:

$$f(2) = (2 - 2)q(2) = \boxed{0}.$$

Why the other options are incorrect: A: -2 , C: 1 , and D: 2 are all nonzero remainders and would mean that $(x - 2)$ is not a factor. The number 2 is the root, not the value of the polynomial at that root.

16. When is the scalar product of two nonzero vectors maximum?

Correct option: A: parallel (same direction)

For fixed magnitudes,

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta.$$

Since $\cos \theta \leq 1$, the maximum is $|\mathbf{a}||\mathbf{b}|$ at $\theta = 0^\circ$: the vectors point in the same direction. **Why the other options are incorrect:** B: opposite directions give $\cos 180^\circ = -1$, the minimum. C: perpendicular vectors give zero. D: at 45° , the product is $|\mathbf{a}||\mathbf{b}|/\sqrt{2}$, smaller than the maximum. **Precision:** the magnitudes are held fixed while the angle varies. Without fixing them, the scalar product has no universal finite upper bound.

Section A | Multiple-choice solutions

Questions 17–20

17. Dot product of $u = 2i + 3j - 4k$ and $v = -2i - k$

Correct option: C: 0

Multiply corresponding components and add. The missing j component of v is zero:

$$u \cdot v = 2(-2) + 3(0) + (-4)(-1) = -4 + 0 + 4 = \boxed{0}.$$

Why the other options are incorrect: A: 6, B: 5 and D: -2 do not equal this sum. In particular, the two k components have the same negative sign, so their product is $+4$. The zero result also shows the vectors are perpendicular.

18. Condition for three vectors to be coplanar

Correct option: D: $P \cdot (Q \times R) = 0$

The absolute value of the scalar triple product is the volume of the parallelepiped formed by the three vectors. They are coplanar exactly when this volume is zero:

$$\boxed{P \cdot (Q \times R) = 0}.$$

Why the other options are incorrect: A: $P \times (Q \times R) = 0$ is a vector triple-product condition, not the coplanarity test. For example, $P = i$, $Q = j$, $R = i$ are coplanar but $P \times (Q \times R) = j \neq 0$. B: $(P \cdot Q) \times R$ attempts a cross product with a scalar, which is undefined. C: $(P \cdot Q) \cdot R$ likewise attempts a dot product between a scalar and a vector, so is not a valid scalar triple product.

19. Expansion of $\cos(A - B)$

Correct option: B: $\cos A \cos B + \sin A \sin B$

Substitute $-B$ in the cosine addition formula, using $\cos(-B) = \cos B$ and $\sin(-B) = -\sin B$:

$$\cos(A - B) = \cos A \cos B - \sin A \sin(-B) = \boxed{\cos A \cos B + \sin A \sin B}.$$

Why the other options are incorrect: A: $\cos A \cos B - \sin A \sin B = \cos(A + B)$. C: $\cos A \sin B + \sin A \cos B = \sin(A + B)$. D: $\cos A \sin B - \sin A \cos B = \sin(B - A)$.

20. Evaluate $\sin 60^\circ \sin 30^\circ$ by product-to-sum

Correct option: A: $\sqrt{3}/4$

Use $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$:

$$\sin 60^\circ \sin 30^\circ = \frac{1}{2}(\cos 30^\circ - \cos 90^\circ) = \frac{1}{2} \left(\frac{\sqrt{3}}{2} - 0 \right) = \boxed{\frac{\sqrt{3}}{4}}.$$

Why the other options are incorrect: B: $\sqrt{3}/2$ omits the factor $1/2$. C: 1 is not $(\sqrt{3}/2)(1/2)$. D: -1 also has the wrong sign because both sines are positive.

Section B | Question 2(i)

Both alternatives | 4 marks each

Main: Sum the first five terms of $\frac{2}{3}, -1, \frac{3}{2}, \dots$

The first term is $a = 2/3$. Check the ratio of consecutive terms:

$$r = \frac{-1}{2/3} = -\frac{3}{2}, \quad \frac{3/2}{-1} = -\frac{3}{2}.$$

The series is geometric. For $r \neq 1$, the sum of its first n terms is

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

Putting $a = 2/3$, $r = -3/2$ and $n = 5$ gives

$$S_5 = \frac{\frac{2}{3}[1 - (-\frac{3}{2})^5]}{1 + \frac{3}{2}} = \frac{\frac{2}{3}(1 + \frac{243}{32})}{\frac{5}{2}} = \frac{2}{3} \cdot \frac{275}{32} \cdot \frac{2}{5} = \frac{55}{24}.$$

Answer: $S_5 = \frac{55}{24}$.

Check by direct addition:

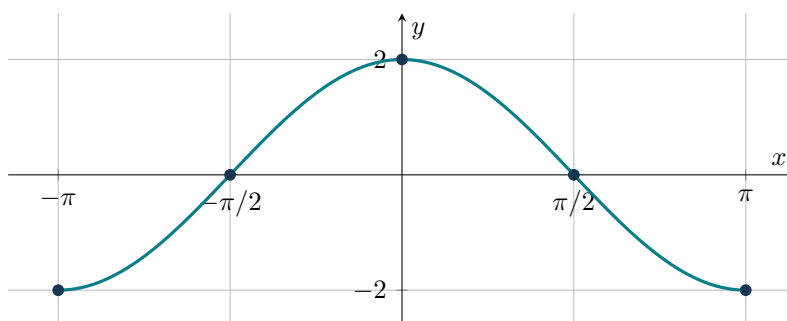
$$\frac{2}{3} - 1 + \frac{3}{2} - \frac{9}{4} + \frac{27}{8} = \frac{16 - 24 + 36 - 54 + 81}{24} = \frac{55}{24}.$$

OR: Sketch $y = 2 \cos x$ **for** $-\pi \leq x \leq \pi$

The cosine curve has amplitude 2 and period 2π . Its important points are:

x	$-\pi$	$-\pi/2$	0	$\pi/2$	π
$y = 2 \cos x$	-2	0	2	0	-2

Plot these points and join them with a smooth cosine curve, symmetric about the y -axis.



Answer: Restricted domain: $[-\pi, \pi]$; range: $[-2, 2]$. The endpoints are included.

Section B | Question 2(ii)

Both alternatives | 4 marks each

Main: Find the last two digits of 12^{11}

The last two digits are the remainder on division by 100. Write $12 = 10 + 2$ and use the binomial theorem:

$$(10 + 2)^{11} = 2^{11} + \binom{11}{1} 10 \cdot 2^{10} + \binom{11}{2} 10^2 \cdot 2^9 + \dots$$

Every term containing 10^2 or a higher power of 10 is divisible by 100 and has no effect on the last two digits. Hence

$$\begin{aligned} 12^{11} &\equiv 2^{11} + 11(10)2^{10} \pmod{100} \\ &\equiv 2048 + 112640 \pmod{100} \\ &\equiv 114688 \equiv 88 \pmod{100}. \end{aligned}$$

Answer: The last two digits of 12^{11} are **88**.

Why this works: numbers differing by a multiple of 100 have the same last two digits. Retaining only the first two terms is valid here because all remaining terms contain the factor 100.

OR: Find the rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$

Use elementary row operations, which preserve rank:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

There are exactly two nonzero rows in echelon form. Thus

Answer: $\text{rank}(A) = 2$.

Independent check: row 2 is twice row 1, so the rank is less than 3. However, the minor using rows 1 and 3, columns 1 and 2, is

$$\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0,$$

so the rank is at least 2. Together these observations prove that the rank is exactly 2.

Section B | Question 2(iii)

Both alternatives | 4 marks each

Main: Multiply two complex numbers in polar form

Given

$$z_1 = 2(\cos 30^\circ + i \sin 30^\circ), \quad z_2 = 3(\cos 60^\circ + i \sin 60^\circ).$$

For complex numbers in polar form, multiply the moduli and add the arguments:

$$r_1(\cos \theta_1 + i \sin \theta_1)r_2(\cos \theta_2 + i \sin \theta_2) = r_1r_2[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)].$$

Therefore,

$$\begin{aligned} z_1 z_2 &= (2)(3)[\cos(30^\circ + 60^\circ) + i \sin(30^\circ + 60^\circ)] \\ &= 6(\cos 90^\circ + i \sin 90^\circ). \end{aligned}$$

Answer: Polar form: $6(\cos 90^\circ + i \sin 90^\circ)$, equivalently $6(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$.

Check: $z_1 = \sqrt{3} + i$, $z_2 = 3/2 + (3\sqrt{3}/2)i$. Multiplying gives real part $3\sqrt{3}/2 - 3\sqrt{3}/2 = 0$ and imaginary part $9/2 + 3/2 = 6$, so $z_1 z_2 = 6i$.

OR: Find n if ${}^n P_4 : {}^{n-1} P_3 = 9 : 1$

The permutation formula is

$${}^n P_r = \frac{n!}{(n-r)!}.$$

Here n must be an integer with $n \geq 4$. Thus

$$\frac{{}^n P_4}{{}^{n-1} P_3} = \frac{n!/(n-4)!}{(n-1)!/(n-4)!} = \frac{n!}{(n-1)!} = n.$$

The given ratio is 9, so

Answer: $n = 9$.

Verification:

$${}^9 P_4 = 9 \cdot 8 \cdot 7 \cdot 6 = 3024, \quad {}^8 P_3 = 8 \cdot 7 \cdot 6 = 336, \quad 3024 : 336 = 9 : 1.$$

Section B | Question 2(iv)

Both alternatives | 4 marks each

Main: Remainder when 9^{205} is divided by 31, using binomial expansion

Use a power whose remainder is simple:

$$9^5 = 59049 = 31(1904) + 25.$$

By the binomial theorem, every term except the last in $(31 \cdot 1904 + 25)^3$ contains a factor 31. Thus

$$9^{15} = (31 \cdot 1904 + 25)^3 \equiv 25^3 = 15625 = 31(504) + 1 \pmod{31}.$$

Hence $9^{15} = 31q + 1$ for some integer q . Since $205 = 15(13) + 10$,

$$9^{205} = (9^{15})^{13} 9^{10} = (31q + 1)^{13} 9^{10}.$$

Expand the first factor:

$$(31q + 1)^{13} = 1 + \binom{13}{1} 31q + \binom{13}{2} (31q)^2 + \cdots + (31q)^{13}.$$

All terms other than 1 are divisible by 31, so $(31q + 1)^{13} \equiv 1 \pmod{31}$. Also,

$$9^{10} = (9^5)^2 = (31 \cdot 1904 + 25)^2 \equiv 25^2 = 625 = 31(20) + 5 \pmod{31}.$$

Combining these results,

$$9^{205} \equiv 1 \cdot 5 \equiv 5 \pmod{31}.$$

Answer: The remainder is 5.

OR: Find all roots of $x^3 - 3x^2 - 4x + 12 = 0$ by the factor theorem

Let $f(x) = x^3 - 3x^2 - 4x + 12$. Test $x = 3$:

$$f(3) = 27 - 27 - 12 + 12 = 0.$$

By the factor theorem, $(x - 3)$ is a factor. Dividing gives $x^2 - 4$, so

$$f(x) = (x - 3)(x^2 - 4) = (x - 3)(x - 2)(x + 2).$$

Set each factor equal to zero:

Answer: All roots are $x = 3, 2, -2$.

Check: expansion of $(x - 3)(x^2 - 4)$ returns $x^3 - 3x^2 - 4x + 12$.

Section B | Question 2(v)

Both alternatives | 4 marks each

Main: Find cofactors $A_{13}, A_{23}, A_{31}, A_{33}$

The given matrix is

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}.$$

For the cofactor A_{ij} , delete row i and column j to form the minor, then multiply its determinant by $(-1)^{i+j}$:

$$A_{ij} = (-1)^{i+j} M_{ij}.$$

Compute the four requested cofactors:

$$A_{13} = (-1)^4 \begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix} = 12 - 12 = 0,$$

$$A_{23} = (-1)^5 \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} = -(6 - 6) = 0,$$

$$A_{31} = (-1)^4 \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0,$$

$$A_{33} = (-1)^6 \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0.$$

Answer: $A_{13} = A_{23} = A_{31} = A_{33} = 0$.

Notation: capital A_{ij} denotes a cofactor here, not the entry a_{ij} . All rows are multiples of the first row, which explains why every 2×2 minor vanishes.

OR: Prove $2^n \geq n + 1$ **for all integers** $n \geq 1$ **by induction**

Base case ($n = 1$): $2^1 = 2$ and $1 + 1 = 2$, so the inequality is true.

Induction hypothesis: assume that, for an arbitrary integer $k \geq 1$,

$$2^k \geq k + 1.$$

Induction step: multiplying by the positive number 2 preserves the inequality:

$$2^{k+1} = 2 \cdot 2^k \geq 2(k + 1) = 2k + 2.$$

Since $k \geq 1$ (and hence $k \geq 0$),

$$2k + 2 \geq k + 2 = (k + 1) + 1.$$

Thus $2^{k+1} \geq (k + 1) + 1$, as required.

Answer: The statement is true at $n = 1$ and passes from k to $k + 1$. Therefore $2^n \geq n + 1$ for every integer $n \geq 1$.

Section B | Question 2(vi)

Both alternatives | 4 marks each

Main: Solve $(2 + i)\omega = 1$, $(3 - i)z + 5\omega = 2 - i$

From the first equation,

$$\omega = \frac{1}{2 + i} = \frac{2 - i}{(2 + i)(2 - i)} = \frac{2 - i}{4 + 1} = \frac{2 - i}{5}.$$

Therefore $5\omega = 2 - i$. Substitute this into the second equation:

$$(3 - i)z + (2 - i) = 2 - i \Rightarrow (3 - i)z = 0.$$

Since $3 - i \neq 0$, $z = 0$.

Answer: $\omega = \frac{2 - i}{5}$ and $z = 0$.

Verification:

$$(2 + i)\frac{2 - i}{5} = \frac{5}{5} = 1, \quad (3 - i)(0) + 5\frac{2 - i}{5} = 2 - i.$$

OR: Moment about the origin of $\mathbf{F} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ acting at $(1, 2, 3)$

The position vector from the origin to the point of application is

$$\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}.$$

The moment is the cross product *position vector crossed with force*:

$$\begin{aligned} \mathbf{M}_O = \mathbf{r} \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix} \\ &= \mathbf{i}[2(1) - 3(-2)] - \mathbf{j}[1(1) - 3(3)] + \mathbf{k}[1(-2) - 2(3)] \\ &= 8\mathbf{i} + 8\mathbf{j} - 8\mathbf{k}. \end{aligned}$$

Answer: $\mathbf{M}_O = 8\mathbf{i} + 8\mathbf{j} - 8\mathbf{k}$. If position is in metres and force in newtons, the units are N m.

Optional magnitude:

$$|\mathbf{M}_O| = \sqrt{8^2 + 8^2 + (-8)^2} = 8\sqrt{3}.$$

Check: a cross product is perpendicular to both vectors:

$$\mathbf{M}_O \cdot \mathbf{r} = 8 + 16 - 24 = 0, \quad \mathbf{M}_O \cdot \mathbf{F} = 24 - 16 - 8 = 0.$$

Reversing the order to $\mathbf{F} \times \mathbf{r}$ would give the negative of the correct moment.

Section B | Question 2(vii)

Both alternatives | 4 marks each

Main: Verify $\cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha$

Start from the left-hand side and apply the cosine addition formula:

$$\begin{aligned}\cos 3\alpha &= \cos(2\alpha + \alpha) \\ &= \cos 2\alpha \cos \alpha - \sin 2\alpha \sin \alpha.\end{aligned}$$

Substitute $\cos 2\alpha = 2\cos^2 \alpha - 1$ and $\sin 2\alpha = 2\sin \alpha \cos \alpha$:

$$\begin{aligned}\cos 3\alpha &= (2\cos^2 \alpha - 1)\cos \alpha - (2\sin \alpha \cos \alpha)\sin \alpha \\ &= 2\cos^3 \alpha - \cos \alpha - 2\sin^2 \alpha \cos \alpha.\end{aligned}$$

Now use $\sin^2 \alpha = 1 - \cos^2 \alpha$:

$$\begin{aligned}\cos 3\alpha &= 2\cos^3 \alpha - \cos \alpha - 2(1 - \cos^2 \alpha)\cos \alpha \\ &= 2\cos^3 \alpha - \cos \alpha - 2\cos \alpha + 2\cos^3 \alpha \\ &= 4\cos^3 \alpha - 3\cos \alpha.\end{aligned}$$

Answer: L.H.S. = R.H.S.; the identity is verified for every real α .

OR: Form a group of 3 from 5 boys and 4 girls

There are $5 + 4 = 9$ students in total. No restriction is given on the numbers of boys or girls. A group is unordered, so

$$\text{Number of groups} = \binom{9}{3} = \frac{9!}{3!6!} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} = 84.$$

Answer: The group can be formed in 84 ways.

Check by disjoint cases:

Boys	Girls	Number of groups
3	0	$\binom{5}{3}\binom{4}{0} = 10$
2	1	$\binom{5}{2}\binom{4}{1} = 40$
1	2	$\binom{5}{1}\binom{4}{2} = 30$
0	3	$\binom{5}{0}\binom{4}{3} = 4$

Total = $10 + 40 + 30 + 4 = 84$. Do not impose an unstated condition such as “at least one girl.”

Section B | Question 2(viii)

Both alternatives | 4 marks each

Main: Volume of a parallelepiped

Given $\mathbf{a} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{c} = \mathbf{i} + \mathbf{k}$, the volume is the absolute scalar triple product:

$$\begin{aligned} V &= |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left| \begin{vmatrix} 1 & 1 & 2 \\ 2 & 1 & -1 \\ 1 & 0 & 1 \end{vmatrix} \right| \\ &= |1(1 \cdot 1 - (-1)0) - 1(2 \cdot 1 - (-1)1) + 2(2 \cdot 0 - 1 \cdot 1)| \\ &= |1 - 3 - 2| = |-4| = 4. \end{aligned}$$

Answer: $V = 4$ cubic units. The absolute value is necessary because geometric volume is nonnegative.

OR: Maximum and minimum of $y = \frac{1}{4 + 5 \cos 3x}$

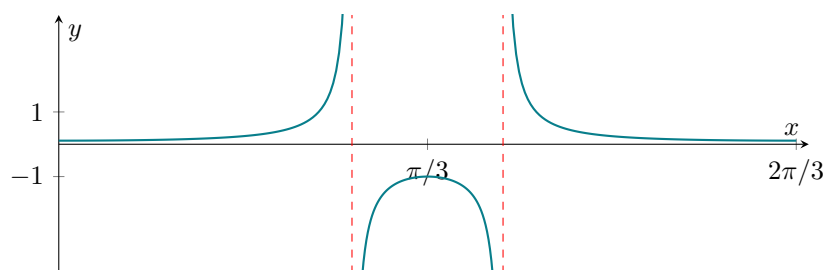
Since $-1 \leq \cos 3x \leq 1$, the denominator $d = 4 + 5 \cos 3x$ takes all values in $[-1, 9]$. In particular, it can be zero. The function is undefined when

$$\cos 3x = -\frac{4}{5}, \quad x = \frac{2k\pi \pm \arccos(-4/5)}{3}, \quad k \in \mathbb{Z}.$$

On the negative-denominator branch $-1 \leq d < 0$, $1/d \in (-\infty, -1]$. On the positive branch $0 < d \leq 9$, $1/d \in [1/9, \infty)$. Hence

Answer: Range = $(-\infty, -1] \cup [1/9, \infty)$. There is **no finite global maximum or minimum**.

As $d \rightarrow 0^+$, $y \rightarrow +\infty$; as $d \rightarrow 0^-$, $y \rightarrow -\infty$. Substituting only $\cos 3x = \pm 1$ therefore misses the unbounded behavior. The value $1/9$ is a local minimum on a positive branch; -1 is a local maximum on a negative branch.



One period; dashed lines are vertical asymptotes.

Section B | Question 2(ix)

Both alternatives | 4 marks each

Main: Is $(x + 2)$ a factor of $f(x) = x^3 + 2x^2 - 4x - 8$?

By the factor theorem, $x + 2 = x - (-2)$ is a factor exactly when $f(-2) = 0$. Substitute $x = -2$:

$$\begin{aligned} f(-2) &= (-2)^3 + 2(-2)^2 - 4(-2) - 8 \\ &= -8 + 8 + 8 - 8 = 0. \end{aligned}$$

Answer: Yes, $(x + 2)$ is a factor because $f(-2) = 0$.

The complete factorization provides a useful check:

$$\begin{aligned} x^3 + 2x^2 - 4x - 8 &= x^2(x + 2) - 4(x + 2) \\ &= (x + 2)(x^2 - 4) \\ &= (x + 2)^2(x - 2). \end{aligned}$$

Thus $x + 2$ occurs twice, although the question asks only whether it is a factor.

OR: Sum the first n terms of $3, 9, 27, 81, \dots$

The first term is $a = 3$ and common ratio is $r = 9/3 = 3$. Let

$$S_n = 3 + 9 + 27 + \dots + 3^n.$$

Multiply by 3:

$$3S_n = 9 + 27 + 81 + \dots + 3^{n+1}.$$

Subtract the first equation from the second. All intermediate terms cancel:

$$3S_n - S_n = 3^{n+1} - 3, \quad 2S_n = 3^{n+1} - 3.$$

Answer: $S_n = \frac{3^{n+1} - 3}{2} = \frac{3(3^n - 1)}{2}.$

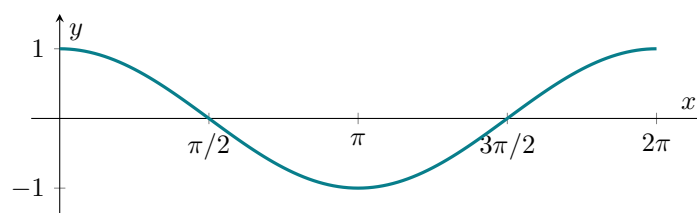
This is also obtained from $S_n = a(r^n - 1)/(r - 1)$. **Check:** for $n = 3$, the formula gives $3(27 - 1)/2 = 39$, while $3 + 9 + 27 = 39$.

Section B | Question 2(x)

Both alternatives | 4 marks each

Main: Domain, range, period and amplitude of $y = \cos x$

Property	Value	Reason
Domain	$\mathbb{R}$	Cosine is defined for every real angle.
Range	$[-1, 1]$	The cosine of an angle lies between -1 and 1 , inclusive.
Fundamental period	2π radians	$\cos(x + 2\pi) = \cos x$; no smaller positive shift repeats the whole curve.
Amplitude	1	Half the difference between maximum and minimum: $(1 - (-1))/2 = 1$.



OR: Find λ for coplanar vectors

The vectors are $\mathbf{v} = (1, -1, 1)$, $\mathbf{u} = (2, 1, -1)$ and $\mathbf{w} = (-1, \lambda, 2)$. Three vectors are coplanar when their scalar triple product is zero:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & \lambda & 2 \end{vmatrix} = 0.$$

Expand along the first row:

$$\begin{aligned} 2[(-1)2 - \lambda] - [1(2) - 1(-1)] + (-1)[\lambda - 1] &= 0, \\ -4 - 2\lambda - 3 - \lambda + 1 &= 0, \\ -6 - 3\lambda &= 0. \end{aligned}$$

Answer: $\lambda = -2$.

Check: with $\lambda = -2$, $\mathbf{w} = (-1, -2, 2) = \mathbf{v} - \mathbf{u}$, a linear combination of the other two vectors. Hence all three lie in the same plane through the origin.

Section B | Question 2(xi)

Both alternatives | 4 marks each

Main: Sum the first 20 terms of an A.P. with $a = 1$, $d = 3$

For an arithmetic progression,

$$S_n = \frac{n}{2}[2a + (n-1)d].$$

Substitute the given values:

$$\begin{aligned} S_{20} &= \frac{20}{2}[2(1) + (20-1)3] \\ &= 10[2 + 57] = 10(59) = 590. \end{aligned}$$

Answer: $S_{20} = 590$.

Check using first and last terms:

$$a_{20} = 1 + 19(3) = 58, \quad S_{20} = \frac{20}{2}(1 + 58) = 590.$$

There are ten pairs, each with sum 59.

OR: Verify $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

Start with the definition of tangent and apply the sine and cosine addition identities:

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} \\ &= \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}. \end{aligned}$$

Divide both numerator and denominator by $\cos \alpha \cos \beta$:

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta}}{1 - \frac{\sin \alpha}{\cos \alpha} \frac{\sin \beta}{\cos \beta}} \\ &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}. \end{aligned}$$

Answer: L.H.S. = R.H.S.; the identity is verified wherever both sides are defined.

Conditions: $\cos \alpha \neq 0$, $\cos \beta \neq 0$ and $1 - \tan \alpha \tan \beta \neq 0$. The last condition ensures $\cos(\alpha + \beta) \neq 0$ when the first two conditions hold.

Section B | Question 2(xii)

Both alternatives | 4 marks each

Main: Find the inverse, if possible, of $A = \begin{pmatrix} 0 & 3 & 4 \\ -4 & 2 & 6 \\ 2 & 0 & 5 \end{pmatrix}$

First compute the determinant by expanding along row 1:

$$|A| = 0 - 3[(-4)(5) - 6(2)] + 4[(-4)(0) - 2(2)] = 96 - 16 = 80.$$

Since $|A| \neq 0$, the inverse exists. Compute all nine cofactors:

$$\begin{aligned} C_{11} &= 2(5) - 6(0) = 10, & C_{12} &= -[(-4)(5) - 6(2)] = 32, \\ C_{13} &= (-4)(0) - 2(2) = -4, & C_{21} &= -[3(5) - 4(0)] = -15, \\ C_{22} &= 0(5) - 4(2) = -8, & C_{23} &= -[0(0) - 3(2)] = 6, \\ C_{31} &= 3(6) - 4(2) = 10, & C_{32} &= -[0(6) - 4(-4)] = -16, \\ C_{33} &= 0(2) - 3(-4) = 12. \end{aligned}$$

Form the cofactor matrix and transpose it:

$$C = \begin{pmatrix} 10 & 32 & -4 \\ -15 & -8 & 6 \\ 10 & -16 & 12 \end{pmatrix}, \quad \text{adj } A = C^T = \begin{pmatrix} 10 & -15 & 10 \\ -16 & -8 & -4 \\ 32 & 6 & 12 \end{pmatrix}.$$

Answer: $A^{-1} = \frac{1}{80} \begin{pmatrix} 10 & -15 & 10 \\ 32 & -8 & -16 \\ -4 & 6 & 12 \end{pmatrix}.$

Verification:

$$A(\text{adj } A) = \begin{pmatrix} 96 - 16 & -24 + 24 & -48 + 48 \\ -40 + 64 - 24 & 60 - 16 + 36 & -40 - 32 + 72 \\ 20 - 20 & -30 + 30 & 20 + 60 \end{pmatrix} = 80I_3.$$

Thus $AA^{-1} = I_3$.

OR: Convert $\cos \frac{2\theta + 2\phi}{2} \sin \frac{2\theta - 2\phi}{2}$ to sum/difference form

Simplify the angles: the expression is $\cos(\theta + \phi) \sin(\theta - \phi)$. Use

$$\sin B \cos A = \frac{1}{2} [\sin(B + A) + \sin(B - A)].$$

With $A = \theta + \phi$ and $B = \theta - \phi$,

$$\begin{aligned} \cos(\theta + \phi) \sin(\theta - \phi) &= \frac{1}{2} [\sin 2\theta + \sin(-2\phi)] \\ &= \frac{1}{2} (\sin 2\theta - \sin 2\phi). \end{aligned}$$

Answer: $\frac{1}{2} (\sin 2\theta - \sin 2\phi).$

Section C | Question 3

Main alternative | 8 marks (4 + 2 + 2)

Given sequence: $4, 2, 1, \frac{1}{2}, \dots$

(a) Identify the sequence and find the general term

The consecutive differences are

$$2 - 4 = -2, \quad 1 - 2 = -1, \quad \frac{1}{2} - 1 = -\frac{1}{2}.$$

These are not equal, so the sequence is not an A.P. However, its consecutive ratios are

$$\frac{2}{4} = \frac{1}{2}, \quad \frac{1}{2} = \frac{1}{2}, \quad \frac{1/2}{1} = \frac{1}{2}.$$

Thus it is a geometric progression with first term $a = 4$ and common ratio $r = 1/2$.

The general term of a G.P. is $a_n = ar^{n-1}$. Consequently,

Answer: The sequence is a G.P., and $a_n = 4 \left(\frac{1}{2}\right)^{n-1} = 2^{3-n}$, $n \geq 1$.

Check: setting $n = 1, 2, 3, 4$ gives $4, 2, 1, 1/2$, respectively.

(b) Find the sum of the first n terms

For $r \neq 1$,

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

Substitute $a = 4$, $r = 1/2$:

$$S_n = \frac{4[1 - (1/2)^n]}{1 - 1/2} = 8 \left(1 - \frac{1}{2^n}\right).$$

Answer: $S_n = 8(1 - 2^{-n})$.

For example, $S_4 = 4 + 2 + 1 + 1/2 = 15/2$; the formula gives $8(1 - 1/16) = 15/2$.

(c) Find the sum to infinity

An infinite geometric series converges only when $|r| < 1$. Here $|r| = 1/2 < 1$, so

$$S_\infty = \frac{a}{1 - r} = \frac{4}{1 - 1/2} = \boxed{8}.$$

Equivalently, $2^{-n} \rightarrow 0$ as $n \rightarrow \infty$, and hence $S_n \rightarrow 8$.

Answer: $S_\infty = 8$.

Exam point: state the convergence condition before applying the infinite-sum formula. The sum approaches 8 as more terms are added; no finite partial sum equals 8.

Section C | Question 3

OR alternative | 8 marks

Select 5 members from 8 men and 6 women, including at least 2 women

The phrase “at least 2 women” allows 2, 3, 4 or 5 women. For each case, select the women and men independently and multiply the two combination counts. Then add the disjoint cases.

Women	Men	Selection formula	Number
2	3	$\binom{6}{2}\binom{8}{3}$	$15 \cdot 56 = 840$
3	2	$\binom{6}{3}\binom{8}{2}$	$20 \cdot 28 = 560$
4	1	$\binom{6}{4}\binom{8}{1}$	$15 \cdot 8 = 120$
5	0	$\binom{6}{5}\binom{8}{0}$	$6 \cdot 1 = 6$

For clarity, the needed combinations are obtained from

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

For instance,

$$\binom{6}{2} = \frac{6 \cdot 5}{2 \cdot 1} = 15, \quad \binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

Therefore,

$$\begin{aligned} N &= \binom{6}{2}\binom{8}{3} + \binom{6}{3}\binom{8}{2} + \binom{6}{4}\binom{8}{1} + \binom{6}{5}\binom{8}{0} \\ &= 840 + 560 + 120 + 6 \\ &= 1526. \end{aligned}$$

Answer: The committee can be formed in 1526 ways.

Independent check: complement method

There are 14 people altogether, so the unrestricted number of committees is $\binom{14}{5} = 2002$. Exclude committees with no women and exactly one woman:

$$N = \binom{14}{5} - \binom{8}{5} - \binom{6}{1}\binom{8}{4} = 2002 - 56 - 6(70) = 1526.$$

Both methods agree. **Common errors to avoid:**

- Counting only two women: “at least two” includes three, four and five women.
- Using permutations: membership is being selected, with no offices or ordering specified.
- Adding the women and men counts within a case: one selection of women is paired with one selection of men, so multiply within a case and add across cases.

Section C | Question 4

Main alternative | 8 marks

Solve the simultaneous linear equations

$$2x + y - z = 1, \quad x + 2y + z = 4, \quad 3x + y + 2z = 7.$$

Use row reduction on the augmented matrix. First interchange rows 1 and 2 to obtain a convenient leading 1:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 1 & 2 & 1 & 4 \\ 3 & 1 & 2 & 7 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 3 & 1 & 2 & 7 \end{array} \right].$$

Eliminate the entries below the first pivot:

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & -5 & -1 & -5 \end{array} \right].$$

Replace R_3 by $3R_3 - 5R_2$ (a nonzero row scaling followed by a row addition):

$$\xrightarrow{R_3 \rightarrow 3R_3 - 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & 0 & 12 & 20 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3/12} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -3 & -7 \\ 0 & 0 & 1 & 5/3 \end{array} \right].$$

Eliminate the z entry from row 2 and normalize that row:

$$\xrightarrow{R_2 \rightarrow R_2 + 3R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & 0 & -2 \\ 0 & 0 & 1 & 5/3 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2/3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 0 & 2/3 \\ 0 & 0 & 1 & 5/3 \end{array} \right].$$

Finally, apply $R_1 \rightarrow R_1 - 2R_2 - R_3$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2/3 \\ 0 & 0 & 1 & 5/3 \end{array} \right].$$

Answer: $x = 1, \quad y = \frac{2}{3}, \quad z = \frac{5}{3}.$

Verification in all three original equations

$$\begin{aligned} 2(1) + \frac{2}{3} - \frac{5}{3} &= 2 - 1 = 1, \\ 1 + 2\left(\frac{2}{3}\right) + \frac{5}{3} &= 1 + 3 = 4, \\ 3(1) + \frac{2}{3} + 2\left(\frac{5}{3}\right) &= 3 + 4 = 7. \end{aligned}$$

All three equations are satisfied. There is a pivot for each unknown, so this solution is unique.

Section C | Question 4

OR alternative | 8 marks

Mechanical arm: $h(t) = 15 \cos(t/2)$ metres

Time t is measured in seconds, $t \geq 0$, and the trigonometric argument is in radians.

First time the arm reaches 7.5 metres

Set the height equal to 7.5:

$$15 \cos\left(\frac{t}{2}\right) = 7.5 \Rightarrow \cos\left(\frac{t}{2}\right) = \frac{1}{2}.$$

Cosine equals $1/2$ at angles $2k\pi \pm \pi/3$. Therefore

$$\frac{t}{2} = 2k\pi \pm \frac{\pi}{3} \Rightarrow t = 4k\pi \pm \frac{2\pi}{3}, \quad k \in \mathbb{Z}.$$

The least nonnegative time satisfying this is

Answer: $t = \frac{2\pi}{3} \text{ s} \approx 2.094 \text{ s}.$

Check: $h(2\pi/3) = 15 \cos(\pi/3) = 15(1/2) = 7.5 \text{ m}.$ The initial height is $h(0) = 15 \text{ m}$, and the curve decreases to this value during the first quarter-period.

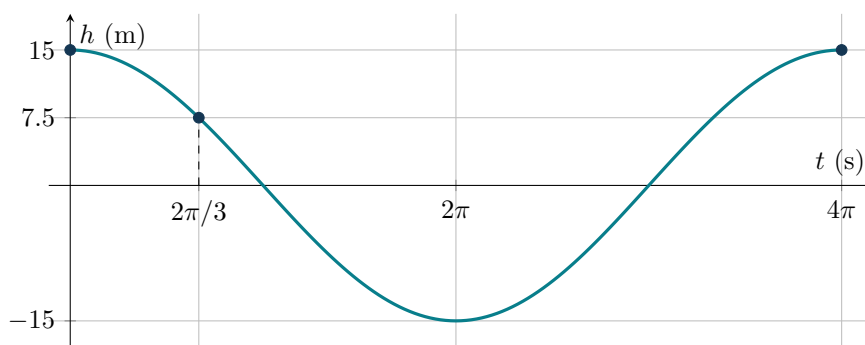
Time for maximum height

Since $\cos(t/2) \leq 1$, the maximum height is 15 m. It occurs when

$$\cos(t/2) = 1 \Rightarrow \frac{t}{2} = 2k\pi \Rightarrow t = 4k\pi.$$

For nonnegative time, $k = 0, 1, 2, \dots$

Answer: Maximum height: 15 m. It occurs at $t = 0, 4\pi, 8\pi, \dots$ seconds. The first subsequent maximum after $t = 0$ is at $4\pi \text{ s} \approx 12.566 \text{ s}.$



The model repeats every 4π seconds.

Section C | Question 5

Main alternative | 8 marks

Middle term and ninth term of $\left(3x^2 - \frac{1}{2x}\right)^{10}$

The expression is defined for $x \neq 0$. For $(a + b)^n$, the $(r + 1)$ th term is

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r, \quad r = 0, 1, \dots, n.$$

Here $a = 3x^2$, $b = -1/(2x)$ and $n = 10$. Thus

$$\begin{aligned} T_{r+1} &= \binom{10}{r} (3x^2)^{10-r} \left(-\frac{1}{2x}\right)^r \\ &= \binom{10}{r} \frac{(-1)^r 3^{10-r}}{2^r} x^{2(10-r)-r} \\ &= \binom{10}{r} \frac{(-1)^r 3^{10-r}}{2^r} x^{20-3r}. \end{aligned}$$

Middle term

There are $10 + 1 = 11$ terms, so there is one middle term: term number $(11 + 1)/2 = 6$. For T_6 , use $r = 5$, not $r = 6$:

$$\begin{aligned} T_6 &= \binom{10}{5} (3x^2)^5 \left(-\frac{1}{2x}\right)^5 \\ &= 252 \cdot 243x^{10} \cdot \left(-\frac{1}{32x^5}\right) \\ &= -\frac{61236}{32}x^5 = -\frac{15309}{8}x^5. \end{aligned}$$

Answer: The only middle term is $T_6 = -\frac{15309}{8}x^5$.

Ninth term

For T_9 , put $r = 8$:

$$\begin{aligned} T_9 &= \binom{10}{8} (3x^2)^2 \left(-\frac{1}{2x}\right)^8 \\ &= 45 \cdot 9x^4 \cdot \frac{1}{256x^8} \\ &= \frac{405}{256x^4}. \end{aligned}$$

Answer: $T_9 = \frac{405}{256x^4}$.

Checks: the general power of x is $20 - 3r$. It gives 5 for $r = 5$ and -4 for $r = 8$. The sixth term is negative because $r = 5$ is odd; the ninth term is positive because $r = 8$ is even.

Section C | Question 5

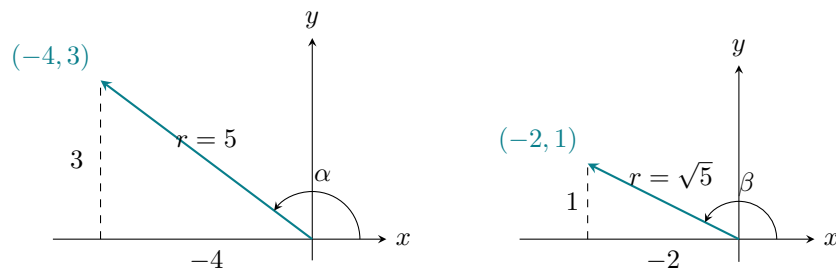
OR alternative | 8 marks

Given $\sin \alpha = \frac{3}{5}$, $\sin \beta = \frac{1}{\sqrt{5}}$, **with both angles in quadrant II**

Find $\cos(\alpha - \beta)$ and $\sin(\alpha + \beta)$.

In quadrant II, sine is positive and cosine is negative. Therefore,

$$\begin{aligned}\cos \alpha &= -\sqrt{1 - \sin^2 \alpha} = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}, \\ \cos \beta &= -\sqrt{1 - \sin^2 \beta} = -\sqrt{1 - \frac{1}{5}} = -\frac{2}{\sqrt{5}}.\end{aligned}$$



Reference triangles establish the negative signs of both cosines.

(i) Evaluate $\cos(\alpha - \beta)$

Use the cosine difference formula:

$$\begin{aligned}\cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ &= \left(-\frac{4}{5}\right) \left(-\frac{2}{\sqrt{5}}\right) + \frac{3}{5} \cdot \frac{1}{\sqrt{5}} \\ &= \frac{8}{5\sqrt{5}} + \frac{3}{5\sqrt{5}} = \frac{11}{5\sqrt{5}} = \frac{11\sqrt{5}}{25}.\end{aligned}$$

Answer: $\cos(\alpha - \beta) = \frac{11\sqrt{5}}{25}$.

(ii) Evaluate $\sin(\alpha + \beta)$

Use the sine addition formula:

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= \frac{3}{5} \left(-\frac{2}{\sqrt{5}}\right) + \left(-\frac{4}{5}\right) \frac{1}{\sqrt{5}} \\ &= -\frac{6}{5\sqrt{5}} - \frac{4}{5\sqrt{5}} = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}.\end{aligned}$$

Answer: $\sin(\alpha + \beta) = -\frac{2\sqrt{5}}{5}$.

Section C | Question 6

Main alternative | 8 marks (3 + 5)

Given polynomial $f(x) = 2x^3 - kx^2 + 16x + 12$

(a) Find k when division by $x - 2$ has zero remainder

The remainder theorem states that the remainder on dividing $f(x)$ by $x - a$ is $f(a)$. Here the zero-remainder condition is $f(2) = 0$:

$$\begin{aligned} 2(2)^3 - k(2)^2 + 16(2) + 12 &= 0, \\ 16 - 4k + 32 + 12 &= 0, \\ 60 - 4k &= 0, \\ k &= 15. \end{aligned}$$

Answer: $k = 15$.

(b) Factorize the polynomial completely

Substitute $k = 15$:

$$f(x) = 2x^3 - 15x^2 + 16x + 12.$$

Since $f(2) = 0$, $(x - 2)$ is a factor. Synthetic division by $x - 2$ gives

$$\begin{array}{r|rrrr} 2 & 2 & -15 & 16 & 12 \\ & & 4 & -22 & -12 \\ \hline & 2 & -11 & -6 & 0 \end{array}$$

Thus

$$f(x) = (x - 2)(2x^2 - 11x - 6).$$

To factor the quadratic, find two numbers with product $2(-6) = -12$ and sum -11 : they are -12 and 1 . Split the middle term:

$$\begin{aligned} 2x^2 - 11x - 6 &= 2x^2 - 12x + x - 6 \\ &= 2x(x - 6) + 1(x - 6) \\ &= (2x + 1)(x - 6). \end{aligned}$$

Answer: $f(x) = (x - 2)(2x + 1)(x - 6)$, for $k = 15$.

The corresponding zeros are $2, -1/2, 6$. **Verification by expansion:**

$$(x - 2)(2x^2 - 11x - 6) = 2x^3 - 11x^2 - 6x - 4x^2 + 22x + 12 = 2x^3 - 15x^2 + 16x + 12.$$

Section C | Question 6

OR alternative | 8 marks (5 + 3)

(a) Prove invariance of the scalar triple product under cyclic permutation

We must prove

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}).$$

Write $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{b} = (b_1, b_2, b_3)$ and $\mathbf{c} = (c_1, c_2, c_3)$. The scalar triple product is

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

To change the row order from $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ to $(\mathbf{b}, \mathbf{c}, \mathbf{a})$, perform two row interchanges:

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}) \longrightarrow (\mathbf{b}, \mathbf{a}, \mathbf{c}) \longrightarrow (\mathbf{b}, \mathbf{c}, \mathbf{a}).$$

Each interchange reverses the determinant's sign. Two interchanges multiply it by $(-1)^2 = 1$, so the determinant is unchanged. Thus

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

Interpreting each determinant as a scalar triple product proves the required identity. This argument also holds when the determinant is zero.

Answer: A cyclic permutation preserves the scalar triple product. Interchanging only two vectors reverses its sign.

(b) Verify using $\mathbf{i}, \mathbf{j}, \mathbf{k}$

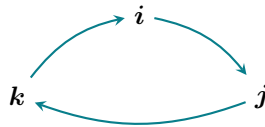
For the standard right-handed orthonormal unit vectors,

$$\mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \quad \mathbf{i} \times \mathbf{j} = \mathbf{k}.$$

Therefore,

$$\begin{aligned} \mathbf{i} \cdot (\mathbf{j} \times \mathbf{k}) &= \mathbf{i} \cdot \mathbf{i} = 1, \\ \mathbf{j} \cdot (\mathbf{k} \times \mathbf{i}) &= \mathbf{j} \cdot \mathbf{j} = 1, \\ \mathbf{k} \cdot (\mathbf{i} \times \mathbf{j}) &= \mathbf{k} \cdot \mathbf{k} = 1. \end{aligned}$$

Answer: All three cyclic scalar triple products equal 1, so the identity is verified.



The cyclic order is $\mathbf{i} \rightarrow \mathbf{j} \rightarrow \mathbf{k} \rightarrow \mathbf{i}$.